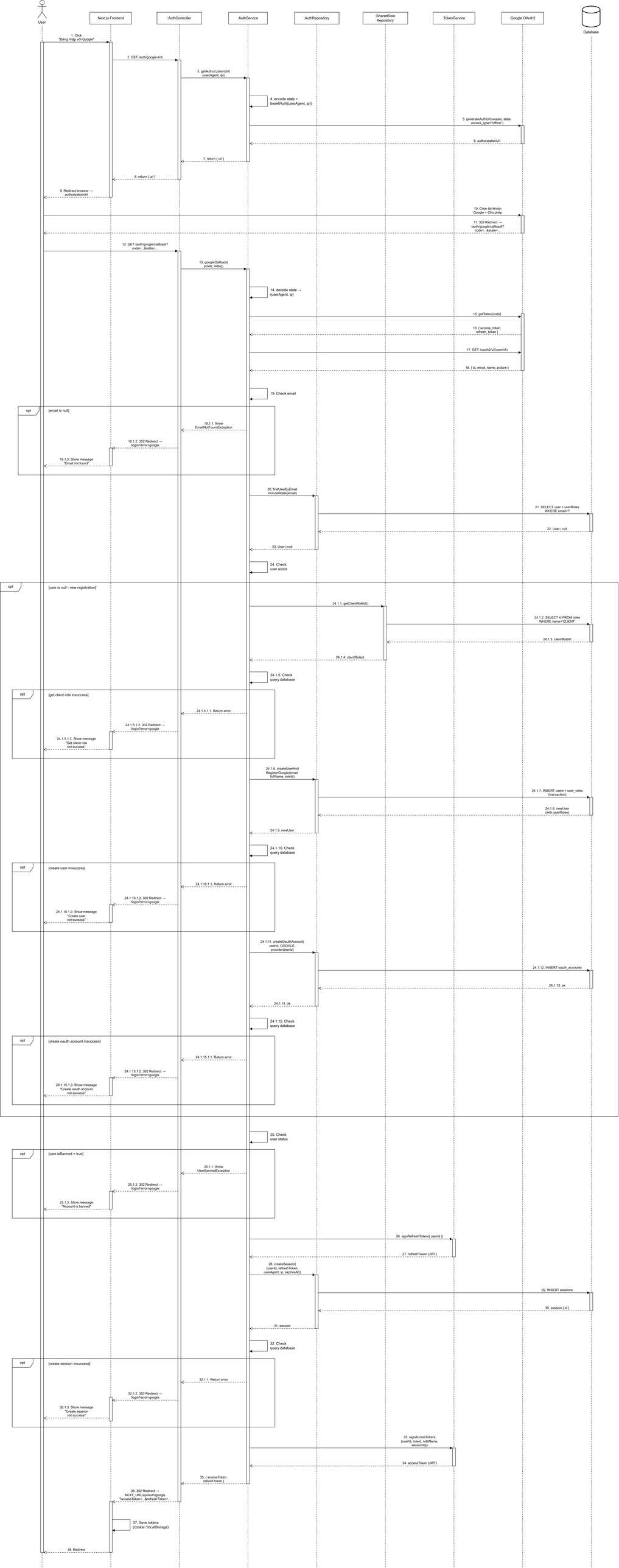

User
:Next.js Frontend
:AuthController
:AuthService
:AuthRepository
:SharedRole Repository
:TokenService
:Google OAuth2
Database
1. Click "Đăng nhập với Google"
2. GET /auth/google-link
3. getAuthorizationUrl({userAgent, ip})
4. encode state = base64url({userAgent, ip})
5. generateAuthUrl({scopes, state, access_type:"offline"})
6. authorizationUrl
7. return { url }
8. return { url }
9. Redirect browser → authorizationUrl
10. Chọn tài khoản Google + Cho phép
11. 302 Redirect → /auth/google/callback? code=...&state=...
12. GET /auth/google/callback? code=...&state=...
13. googleCallback({code, state})
14. decode state → {userAgent, ip}
15. getToken(code)
16. { access_token, refresh_token }
17. GET /oauth2/v2/userinfo
18. { id, email, name, picture }
19. Check email
opt
[email is null]
19.1.1. throw EmailNotFoundException
19.1.2. 302 Redirect → /login?error=google
19.1.3. Show message "Email not found"
20. findUserByEmail IncludeRoles(email)
21. SELECT user + userRoles WHERE email=?
22. User | null
23. User | null
24. Check user exists
opt
[user is null - new registration]
24.1.1. getClientRoleId()
24.1.2. SELECT id FROM roles WHERE name='CLIENT'
24.1.3. clientRoleId
24.1.4. clientRoleId
24.1.5. Check query database
opt
[get client role insuccess]
24.1.5.1.1. Return error
24.1.5.1.2. 302 Redirect → /login?error=google
24.1.5.1.3. Show message "Get client role not success"
24.1.6. createUserAnd RegisterGoogle(email, fullName, roleId)
24.1.7. INSERT users + user_roles (transaction)
24.1.8. newUser (with userRoles)
24.1.9. newUser
24.1.10. Check query database
opt
[create user insuccess]
24.1.10.1.1. Return error
24.1.10.1.2. 302 Redirect → /login?error=google
24.1.10.1.3. Show message "Create user not success"
24.1.11. createOauthAccount(userId, GOOGLE, providerUserId)
24.1.12. INSERT oauth_accounts
24.1.13. ok
24.1.14. ok
24.1.15. Check query database
opt
[create oauth account insuccess]
24.1.15.1.1. Return error
24.1.15.1.2. 302 Redirect → /login?error=google
24.1.15.1.3. Show message "Create oauth account not success"
25. Check user status
opt
[user.isBanned = true]
25.1.1. throw UserBannedException
25.1.2. 302 Redirect → /login?error=google
25.1.3. Show message "Account is banned"
26. signRefreshToken({ userId })
27. refreshToken (JWT)
28. createSession({userId, refreshToken, userAgent, ip, expiresAt})
29. INSERT sessions
30. session { id }
31. session
32. Check query database
opt
[create session insuccess]
32.1.1. Return error
32.1.2. 302 Redirect → /login?error=google
32.1.3. Show message "Create session not success"
33. signAccessToken({userId, roleId, roleName, sessionId})
34. accessToken (JWT)
35. { accessToken, refreshToken }
36. 302 Redirect → NEXT_URL/api/auth/google ?accessToken=...&refreshToken=...
37. Save tokens (cookie / localStorage)
38. Redirect